

English for Technical Communication

A complete Listening, Speaking, Reading and Writing handbook for academic, workplace and professional communication—with concise Malayalam concept support.

Copyright-safe approach: The handbook explains every official skill using original practice passages and an original poem. It does not reproduce unpublished SITTR course-book texts or copyrighted works.

Course at a glance

Semester	I
Teaching Scheme	2:0:2:0
Contact Hours	60
Credits	3
CIE	40
ESE	60
Core Skills	LSRW
Modules	4

Purpose, outcomes and assessment identity

This course connects technical knowledge with professional practice. Students learn to understand information, speak confidently, read technical and non-technical texts critically, and produce clear professional documents.

CO1

Listen and identify main ideas and supporting details.

CO2

Use language functions confidently in workplace and daily contexts.

CO3

Read, analyse, translate and summarise varied texts.

CO4

Plan, draft and produce clear technical and professional documents.

Official course objectives

1. Develop effective listening and comprehension in formal and informal contexts.
2. Build spoken fluency and confidence for academic and workplace situations.
3. Apply reading strategies to technical and non-technical texts.
4. Write accurately, concisely and clearly in technical, professional and creative forms.

Prerequisite and study method

Prerequisite: Basic Listening, Speaking, Reading and Writing skills at Secondary Education level.

Study cycle: Input → Notice language → Practise → Receive feedback → Revise → Apply in a real context.

Malayalam explanation: ് course-ഈ ് ് grammar ് ്. Listening, Speaking, Reading, Writing ് skills real-life, academic, workplace situations-ഈ ് ്.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Total hours	Credits	Theory CIE	Theory ESE	Practical CIE	Total
2	0	2	0	4	60	3	20	60	20	100

CO-PO Mapping

CO	PO2	PO3	PO7	PO8	PO9	PO11
CO1	3	-	-	-	3	2
CO2	3	2	-	2	3	2
CO3	3	2	-	-	3	2

CO	PO2	PO3	PO7	PO8	PO9	PO11
CO4	3	2	2	2	3	2

3 = High, 2 = Medium, 1 = Low, - = No correlation.

LEARNING ROADMAP

From understanding messages to professional communication

Module I

Prose, comprehension, summary, word formation, sentence structure and Situational English.

Module II

Poetry, Tense, Active/Passive Voice, description and Reported Speech.

Module III

Phonetics, stress, intonation, pauses, punctuation, self-introduction, listening and notes.

Module IV

Translation, responsible AI use, email, letter, report and presentation.

Module I — Prose

CO1, CO2, CO4

Theory + Practice: 16 hours

Comprehension · Summary · Sentence Use

1. Listening and Reading Comprehension

Original practice story: The Last Fault Before Launch

On the morning a student-built solar dryer was to be demonstrated, the control display suddenly went blank. Everyone looked at the circuit board, but Meera first checked the simplest point: the emergency-stop button had been pressed during cleaning. She did not reset it immediately. She informed the instructor, isolated the supply and confirmed that no one was touching the wiring. After permission, the team reset the switch and restarted the system. The display returned, but Meera noticed an unusual temperature value. Instead of ignoring it to meet the schedule, she compared the sensor reading with a thermometer. A loose sensor connector was found and secured. The demonstration began ten minutes late, yet it ran safely and accurately. The instructor told the class that technical skill is not only the ability to repair a fault. It is also the discipline to communicate, verify and refuse unsafe shortcuts.

Comprehension check

1. Why did Meera avoid resetting the emergency stop immediately?
2. What evidence showed that the sensor reading was doubtful?
3. Which qualities made Meera's response professional?
4. State the main idea in one sentence.

Model response: Meera followed safe communication and verification procedures instead of choosing a quick but risky solution.

Malayalam explanation: Comprehension പഠിക്കുന്നതിനുള്ള main idea പഠിക്കുന്നതിനുള്ള. പഠിക്കുന്നതിനുള്ള supporting details, sequence, reason, result പഠിക്കുന്നതിനുള്ള. Answer passage-ൽ പഠിക്കുന്നതിനുള്ള copy പഠിക്കുന്നതിനുള്ള പഠിക്കുന്നതിനുള്ള words-ൽ പഠിക്കുന്നതിനുള്ള പഠിക്കുന്നതിനുള്ള.

2. Reading Strategies and Meaning from Context

Skimming

Read title, headings, first/last sentences and repeated keywords to understand the general idea.

Scanning

Move quickly through the text to locate a particular name, date, measurement, instruction or term.

Close reading

Examine argument, evidence, sequence, tone and relationships between ideas.

Context clues

Use definition, example, contrast and cause-effect clues to infer unfamiliar words.

Technical reading tip: Never guess a safety instruction from context alone. Confirm critical terminology in an authorised manual or glossary.

Malayalam explanation: Skimming = overall idea ഉൾക്കൊള്ളൽ. Scanning = specific detail തിരയൽ. Technical safety information ഉൾക്കൊള്ളൽ guess തിരയൽ authorised source ഉൾക്കൊള്ളൽ verify ഉൾക്കൊള്ളൽ.

3. Summarising a Passage

Read twice →

Underline main idea →

Group supporting points →

Remove examples/repetition →

Write in own words →

Check length and meaning

Good summary

- Retains central meaning
- Uses concise original wording
- Follows logical order
- Excludes personal opinion unless asked

Avoid

- Copying long phrases
- Adding new facts
- Including every example
- Changing the writer's meaning

Paste or type a practice summary

Write an 80-100 word summary of the practice story.

Word-count feedback appears here.

Malayalam explanation: Summary ഉൾക്കൊള്ളൽ passage-ഉൾക്കൊള്ളൽ main meaning ഉൾക്കൊള്ളൽ. Examples, repetition ഉൾക്കൊള്ളൽ main points logical order-ഉൾക്കൊള്ളൽ words-ഉൾക്കൊള്ളൽ.

4. Word Formation and Vocabulary

Process	Examples	Effect
Prefix	safe → unsafe; connect → reconnect	Changes meaning
Suffix	develop → development; use → useful	Often changes word class
Synonym	accurate / precise	Similar meaning
Antonym	permanent / temporary	Opposite meaning
Homonym	current (flow / present time)	Same form, different meaning

Word-building practice

connect

re-

Choose a root and affix.

5. Sentence Structure, Articles and Auxiliaries

Pattern	Example
S + V	The alarm rang.
S + V + O	The technician checked the panel.
S + V + C	The reading is stable.
S + V + IO + DO	The supervisor gave the trainee a checklist.
S + V + O + C	The team found the cable damaged.

Articles

- **a** before a consonant sound: a university, a motor.
- **an** before a vowel sound: an hour, an engineer.
- **the** for a specific/unique item: the main switch, the sun.

Auxiliaries

be, have, do form tenses, negatives and questions. Modal auxiliaries such as **can, could, may, must, should** express ability, permission, possibility, obligation and advice.

Malayalam explanation: Article ഏകപദപദങ്ങൾ first letter ഏകപദപദങ്ങൾ; word-പദപദങ്ങൾ sound ഏകപദപദങ്ങൾ. ഏകപദപദങ്ങൾ “an hour”, ഏകപദപദങ്ങൾ “a university” ഏകപദപദങ്ങൾ.

6. Concord / Subject-Verb Agreement

Rule	Correct example
Singular subject → singular verb	The machine works.
Plural subject → plural verb	The machines work.
Each/every/either/neither → singular	Each student has an ID card.
Either...or / neither...nor agrees with nearer subject	Neither the teacher nor the students are late.
Uncountable nouns normally singular	The equipment is ready.
A number of / The number of	A number of students are present. The number is thirty.

Common error: “The equipments are...” is wrong in standard English. Use “The equipment is...” or “The pieces of equipment are...”.

7. Situational English

Request

Could you please repeat the instruction?

Would you mind checking this reading?

Permission

May I enter the laboratory?

Could I use the multimeter?

Clarification

Do you mean that the supply must remain isolated?

Could you explain the second step?

Opinion and reason

In my view, the test should be repeated because the readings are inconsistent.

Workplace dialogue:

Technician: Good morning. Could you confirm whether the panel is isolated?

Supervisor: Yes. The main supply is locked out, but verify it with the tester before starting.

Technician: Certainly. I will record the result in the checklist.

Malayalam explanation: Situational English-☐ sentence grammatically correct ☐☐☐☐☐☐☐☐☐☐, context-☐☐ appropriate, polite, clear ☐☐☐☐☐☐☐☐.

Module I — One-mark Questions

1. What is the main idea of a passage?

1 mark C01

The central message that connects all important details.

2. What is a supporting detail? 1 mark C01

A fact, example, reason or explanation that develops the main idea.

3. Define skimming. 1 mark C03

Reading quickly to understand the general idea and organisation.

4. Define scanning. 1 mark C03

Searching a text quickly for a specific fact, name, number or keyword.

5. What is a summary? 1 mark C04

A brief restatement of the main ideas in the writer's own words.

6. Give one prefix that forms an opposite. 1 mark C02

un-, as in *safe* → *unsafe*.

7. Give one suffix that forms a noun.

1 mark C02

-ment, as in *develop* → *development*.

8. What are synonyms? 1 mark C02

Words with the same or nearly the same meaning.

9. What are antonyms? 1 mark C02

Words with opposite meanings.

10. What are homonyms? 1 mark C02

Words that share spelling or pronunciation but have different meanings.

11. Name the subject in: 'The technician checks the panel.' 1 mark C02

The technician.

12. Choose the article: ___ honest engineer. 1 mark C02

An, because *honest* begins with a vowel sound.

13. Choose the correct verb: Each student ___ a notebook. 1 mark C02

has.

14. What is an auxiliary verb? 1 mark C02

A helping verb used with a main verb to express tense, voice, possibility, obligation or question form.

15. Give one polite request. 1 mark C02

Could you please explain the procedure again?

Module I — Three-mark Questions

1. Differentiate skimming and scanning.

3 marks C01

Skimming finds the general idea; scanning locates a specific detail. Skimming follows headings and topic sentences, while scanning searches for keywords, dates or numbers.

2. Write three rules for an effective summary.

3 marks C04

Retain the main idea, remove repetition/examples, and use original concise sentences without changing the meaning.

3. Explain prefix, suffix and root with one example each.

3 marks C02

A root carries core meaning (*act*); a prefix comes before it (*re+act*); a suffix comes after it (*act+ion*).

4. State three rules of subject-verb agreement.

3 marks C02

A singular subject takes a singular verb; a plural subject takes a plural verb; each/every/either/neither normally takes a singular verb.

5. Explain the use of *a*, *an* and *the*.

3 marks C02

Use *a* before a consonant sound, *an* before a vowel sound, and *the* for a specific or previously identified noun.

6. Write a short workplace conversation requesting permission.

3 marks C02

Employee: May I inspect the control panel now? Supervisor: Yes, but isolate the supply first. Employee: Certainly. I will follow the lockout procedure.

7. Identify the sentence pattern: 'The manager gave the trainee a task.'

3 marks

C02

S + V + IO + DO: The manager (S), gave (V), the trainee (indirect object), a task (direct object).

8. Explain how context clues help vocabulary learning.

3 marks C03

Definitions, examples, contrast words and cause-effect relations around an unfamiliar word help the reader infer its likely meaning.

9. Write three expressions for stating an opinion politely.

3 marks C02

In my view...; I believe that...; From my experience...

10. Write three qualities of effective situational English.

3 marks C02

It should be clear, polite and appropriate to the listener, purpose and context.

Module I — Seven-mark Questions

1. Read an unfamiliar technical passage and explain a complete comprehension strategy. 7 marks CO1

Model points: preview title and headings; predict purpose; read for main idea; mark keywords; infer unfamiliar words from context; identify supporting details; answer in own words; review evidence.

2. Summarise the original practice story in 80-100 words. 7 marks CO4

A strong answer identifies the workshop fault, the trainee's careful observation, the team's safe response and the lesson that disciplined communication prevents accidents. It must be concise, chronological and written in original language.

3. Explain sentence structure, articles, auxiliaries and concord with examples. 7 marks CO2

Award points for sentence elements/patterns, article choice by sound/specificity, functions of auxiliaries and at least three subject-verb agreement rules with correct examples.

4. Prepare a real-life conversation between a student and laboratory instructor. 7 marks CO2

The dialogue should include greeting, purpose, polite request, clarification, confirmation, thanks and appropriate technical vocabulary.

5. Describe the process of forming new words using prefixes and suffixes. 7 marks CO2

Define root/prefix/suffix, explain meaning and word-class changes, give at least six correct examples and caution that spelling may change.

6. Explain how listening and reading support technical learning. 7 marks CO1

Cover instructions, safety information, manuals, lectures, main ideas, details, note-taking, vocabulary and error prevention, with relevant examples.

Module I — MCQ Practice

1. Which strategy is best for locating a serial number? MCQ CO1

Scanning.

2. Which word is a synonym of 'accurate'? MCQ CO2

Precise.

3. Which is correct: 'Each tools are checked' or 'Each tool is checked'? MCQ CO2

Each tool is checked.

4. Choose the correct article before 'hour'. MCQ CO2

An.

5. A concise restatement of main ideas is a ____. MCQ CO4

Summary.

6. Which prefix means 'again'? MCQ CO2

re-.

7. Which sentence is a polite request?

MCQ CO2

Could you please send the report?

8. In SVO, O means ____. MCQ CO2

Object.

9. The opposite of 'temporary' is ____.

MCQ CO2

Permanent.

10. The helping verb in 'The motor is running' is ____. MCQ CO2

is.

Module II — Poetry

CO2, CO3, CO4

Theory + Practice: 15 hours

Appreciation · Tense · Voice · Description

1. Comprehension and Appreciation of Poetry

Blueprint at Dawn

Before the workshop wakes to sound,
A quiet plan lies spread around;
Each measured line, each noted sign,
Turns thought to form through clear design.

The hammer waits, the circuits sleep,
While patient checks their promise keep;
No hurried hand, no careless start—
Safe work begins within the heart.

Then motors hum and screens grow bright,
Ideas become a working light;
Yet skill is more than speed or art:
It speaks, verifies, and does its part.

Original practice poem created for this handbook; not an official SITTR course-book poem.

Literal meaning

A workshop follows planning, checking and safe procedures before productive work begins.

Theme

Real technical skill combines design, patience, safety, communication and verification.

Devices

Personification: “circuits sleep”; metaphor: “working light”; imagery: measured lines, motors and screens.

Tone and message

Calm, respectful and encouraging. The poem values disciplined work over haste.

Read aloud →

Find literal meaning →

Identify speaker/theme →

Observe imagery and sound →

Support response with lines

Malayalam explanation: Poem appreciation- summary പരസ്യം പരസ്യം. Theme, tone, imagery, poetic devices, message പരസ്യം poem-പരസ്യം evidence പരസ്യം

2. Tense Bank

Form	Main use	Technical example
Simple Present	Facts, routines	The sensor measures temperature.
Present Continuous	Action happening now	The technician is testing the relay.
Present Perfect	Past action linked to present	The team has completed the inspection.
Simple Past	Completed past event	The alarm stopped at 10:15.
Past Continuous	Ongoing past action	The motor was running when the fault occurred.
Past Perfect	Earlier of two past actions	The operator had isolated the supply before maintenance began.
Simple Future	Prediction/decision	The engineer will submit the report tomorrow.
Future Continuous	Ongoing future action	The team will be testing the prototype at noon.

Time expression check: Tense must match time and sequence. Do not use Present Perfect with a finished past time such as “yesterday”.

Malayalam explanation: Tense □□□□□□□□□□□□□□□□ verb form □□□□□□ □□□□□□□□□□. Action □□□□□□ □□□□□, duration, sequence, present relevance □□□□□□□□□□ □□□□□□□□.

3. Active and Passive Voice

Active Voice

The doer is the subject.

The technician tested the circuit.

Passive Voice

The receiver/result becomes the subject.

The circuit was tested by the technician.

Find subject-verb-object →

Move object to subject →

Choose correct be form →

Use past participle →

Add agent only if useful

Tense	Active	Passive
Simple Present	They inspect the lift.	The lift is inspected.
Simple Past	They replaced the fuse.	The fuse was replaced.
Present Perfect	They have completed the test.	The test has been completed.
Modal	We must follow the procedure.	The procedure must be followed.

Malayalam explanation: Passive Voice process, result, object എംഫാസൈസ് technical writing-ന് അനുയോജ്യമാണ്. Agent "by..." എംഫാസൈസ്.

4. Describing People, Places, Things and Processes

Person

Identity/role → appearance where relevant → qualities → skills/actions → impression.

Place

General view → spatial order → important features → atmosphere/purpose.

Thing

Name/purpose → size/material/parts → operation → applications.

Process

Purpose/input → sequence → checks/safety → output/result.

Model process description: Testing a continuity path

First, isolate the circuit from all power sources. Next, select the continuity mode on the multimeter and verify the meter using the probes. Place one probe at each end of the conductor. If the path is continuous, the meter produces a tone or displays low resistance. Finally, record the result and restore the equipment only after completing all checks.

Safety: Never perform a continuity test on an energised circuit.

5. Reported Speech

Direct Speech	Reported Speech
Ravi said, "I am ready."	Ravi said that he was ready.
She said, "I will call tomorrow."	She said that she would call the next day.
He asked, "Is the supply off?"	He asked whether the supply was off.
The supervisor said, "Wear your helmet."	The supervisor instructed us to wear our helmets.

Common changes

- Pronouns change according to speaker and listener.
- Present forms may shift to past when reporting later.
- now → then; today → that day; tomorrow → the next day; here → there.
- Reported questions use statement word order.

- Commands use *told/asked/instructed* + *object* + *to-infinitive*.

Malayalam explanation: Reported Speech- words mechanically. Speaker, listener, time, place, reporting context pronoun, tense, time words.

Module II — One-mark Questions

1. What is the theme of a poem? 1 mark

C03

Its central idea or underlying message.

2. Define imagery. 1 mark C03

Language that creates sensory pictures or experiences.

3. What is a simile? 1 mark C03

A comparison using *like* or *as*.

4. What is a metaphor? 1 mark C03

A direct comparison that says one thing is another.

5. What is personification? 1 mark C03

Giving human qualities to a non-human thing.

6. What is alliteration? 1 mark C03

Repetition of initial consonant sounds in nearby words.

7. Which tense expresses a completed past action? 1 mark C02

Simple Past.

8. Which tense is used for an action happening now? 1 mark C02

Present Continuous.

9. Change to passive: 'The operator starts the machine.' 1 mark C04

The machine is started by the operator.

10. Change to active: 'The report was prepared by Maya.' 1 mark C04

Maya prepared the report.

11. What is reported speech? 1 mark C04

Reporting another person's words without quoting them exactly.

12. Report: He said, 'I am ready.' 1 mark C04

He said that he was ready.

13. Name one organising principle for process description. 1 mark C02

Chronological sequence.

14. Give one adjective suitable for describing a workplace. 1 mark C02

Well-organised / safe / spacious.

15. What is tone in poetry? 1 mark C03

The speaker's attitude towards the subject or audience.

Module II — Three-mark Questions

1. Differentiate theme and summary of a poem. 3 marks C03

Theme is the underlying idea; summary states what the poem says or does in sequence.

2. Explain any three poetic devices.

3 marks C03

Example answer: imagery appeals to senses; simile compares using like/as; personification gives human qualities to objects.

3. Write the three principal forms of the Present Tense with examples. 3 marks C02

Simple Present: The motor runs. Present Continuous: The motor is running. Present Perfect: The motor has run for two hours.

4. State three rules for changing Active Voice to Passive Voice. 3 marks C04

Move object to subject position; use suitable form of *be* + past participle; retain tense and add agent only when necessary.

5. Describe a person using three organised points. 3 marks C02

Identify role/appearance, describe qualities and skills, and conclude with impact or relationship.

6. Describe a place using spatial order.

3 marks C02

Start with general view, move left-to-right or near-to-far, mention key features and end with overall impression.

7. Write three sequence markers used in process description. 3 marks C02

First, next, finally.

8. State three changes commonly made in Reported Speech. 3 marks C04

Pronouns change by context, tense may shift back, and time/place words may change.

9. Convert into Reported Speech: She said, 'I will submit the report tomorrow.'

3 marks C04

She said that she would submit the report the next day.

10. Explain how to appreciate a poem.

3 marks C03

Read for literal meaning, identify speaker/theme/tone, note imagery and figures of speech, observe sound/rhythm, and state personal response with evidence.

Module II — Seven-mark Questions

1. Appreciate the original poem 'Blueprint at Dawn'. 7 marks C03

Cover title, central idea of disciplined work, tone, imagery, metaphor/personification, sound effects, message and personal response with evidence.

2. Explain the complete tense system with suitable technical examples. 7 marks C02

Organise Past, Present and Future; explain Simple, Continuous, Perfect and Perfect Continuous uses with accurate examples.

3. Explain Active and Passive Voice and convert five sentences. 7 marks CO4

Define both voices, give transformation method, preserve tense, explain agent omission and supply five correct conversions.

4. Describe a technical process clearly. 7 marks CO2

Use title/purpose, required inputs, chronological steps, sequence markers, Passive Voice where suitable, safety note and result.

5. Explain Reported Speech with statements, questions and commands. 7 marks CO4

Cover reporting verbs, tense/pronoun/time changes, question word order and infinitive form for commands, with examples.

6. Compare description of a person, place, object and process. 7 marks CO2

Discuss focus, organisation, vocabulary and sample structure for each type.

Module II – MCQ Practice

1. A comparison using 'like' is a _____. MCQ CO3

Simile.

2. 'The machine coughed' is _____. MCQ CO3

Personification.

3. 'She has completed the task' is in _____. MCQ CO2

Present Perfect.

4. The passive form of 'They inspect the lift' begins with _____. MCQ CO4

The lift is inspected...

5. A process is best organised in _____. MCQ CO2

Chronological order.

6. The Reported Speech of 'I can help' normally uses _____. MCQ CO4

could.

7. The feeling created by a poem is its _____. MCQ CO3

Mood.

8. Repeated initial sounds form _____. MCQ CO3

Alliteration.

9. 'First, then, finally' are _____. MCQ CO2

Sequence markers.

10. A direct quotation requires _____. MCQ CO4

Quotation marks.

Module III — Listening and Speaking

1. Phonetics and Speech Sounds

Phonetics studies how speech sounds are produced and heard. English spelling does not always show pronunciation accurately, so learners use the International Phonetic Alphabet (IPA) as a sound guide.

Contrast	Minimal pair	Focus
/ɪ/ vs /i:/	ship / sheep	Short and long vowel
/p/ vs /b/	pat / bat	Voiceless and voiced
/f/ vs /v/	fan / van	Voice contrast
/s/ vs /ʃ/	sip / ship	Tongue and airflow
/t/ vs /θ/	tin / thin	Dental fricative

Pronunciation principle: Aim for intelligibility, not imitation of a particular accent. Clear sounds, stress and phrasing matter more than accent elimination.

Malayalam explanation: Phonetics ശബ്ദങ്ങളുടെ ഉത്പത്തിയും കേൾക്കലും accent ശബ്ദങ്ങളുടെ ഉച്ചാരണവും. Listener-ക്ക് easily understand ശബ്ദങ്ങളുടെ clear pronunciation.

2. Stress, Intonation and Pauses

Word stress

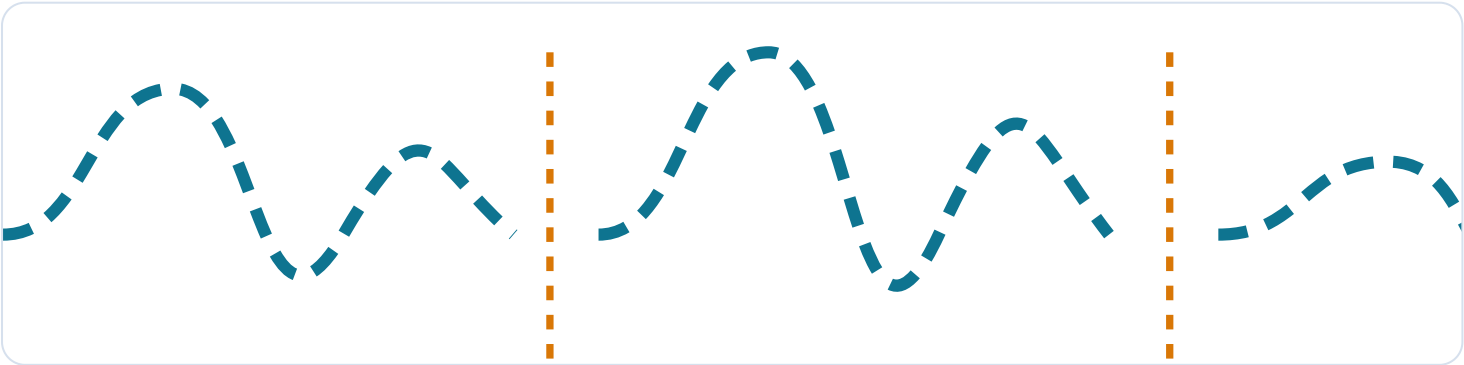
One syllable is more prominent: **PRE**sent (noun), pre**SENT** (verb).

Sentence stress

Content words normally receive stress: **CHECK** the **MAIN SWITCH**.

Intonation

Falling for statements/WH-questions; rising often for yes/no questions; fall-rise for reservation.



Mark idea groups with slashes: *Before starting the test / isolate the power / and verify zero voltage.*

Malayalam explanation: Stress important word- highlight . Intonation voice pitch- rise/fall . Pause idea groups meaning clear .

3. Capital Letters and Punctuation

Mark	Main use	Example
.	End a statement	The test is complete.
,	Separate items/introductory clause	After inspection, record the value.
:	Introduce list/explanation	Bring three items: gloves, goggles and a checklist.
;	Join closely related independent clauses	The reading was unstable; the test was repeated.
?	End a direct question	Is the supply isolated?
" "	Show direct quotation	She said, "Stop the machine."
'	Possession/omission	the engineer's report; don't

Capitalization

Capitalize sentence beginnings, the pronoun I, names, official titles when used with names, organisations, days/months and recognised abbreviations.

Punctuation practice

after testing the motor rina said it is safe

Try correcting it first.

4. Professional Self-introduction

Greeting →

Name and place →

Qualification →

Technical interests →

Strength with evidence →

Goal →

Thanks

Good morning. My name is Anjali S. I am from Kozhikode and I am pursuing a diploma in Electrical and Electronics Engineering. I am interested in control systems and technical drawing. During my course, I have developed strengths in careful documentation, teamwork and systematic problem solving. My immediate goal is to improve my practical skills and contribute effectively during industrial training. Thank you for the opportunity to introduce myself.

Avoid: family biography, unsupported claims such as “I am perfect,” memorised robotic delivery, excessive personal details and negative comments.

Malayalam explanation: Self-introduction short, relevant, confident തിരുത്തലില്ല. Strength തിരുത്തലില്ല തിരുത്തലില്ല evidence തിരുത്തലില്ല. Job/academic context-തിരുത്തലില്ല തിരുത്തലില്ല personal details തിരുത്തലില്ല.

5. Listening and Comprehension

Original listening script: Workshop Safety Briefing

Before entering the test area, wear safety shoes and eye protection. Keep bags outside the marked zone. The equipment will remain isolated until the instructor gives permission. During the demonstration, observe the indicator lamps but do not touch the enclosure. If you hear the warning buzzer, step behind the yellow line and wait for further instructions. After the test, record the voltage, current and time in the observation sheet. Report any unusual smell, vibration or sound immediately.

Questions

1. Where must bags be kept?
2. What should students do when the warning buzzer sounds?
3. Which three readings must be recorded?
4. Name three unusual conditions that must be reported.

Outside the marked zone; move behind the yellow line and wait; voltage/current/time; unusual smell/vibration/sound.

6. Note Making

Identify purpose →

Find main headings →

Select keywords →

Use numbering and indentation →

Add symbols/abbreviations →

Write a key

Notes

Workshop Safety

1. PPE
 - 1.1 safety shoes
 - 1.2 eye protection
2. Test zone
 - 2.1 bags outside
 - 2.2 no touch enclosure
3. Warning buzzer → behind yellow line
4. Record: V, I, time
5. Report unusual smell/vibration/sound

Key

PPE = Personal Protective Equipment

V = Voltage

I = Current

→ = leads to / action

Note making organises information; it is not a full paragraph.

Malayalam explanation: Note Making-□ full sentences □□□□□□□□□□. Headings, subheadings, keywords, numbers, symbols, abbreviations □□□□□□□□□□ information logical structure-□ arrange □□□□□□□□.

Module III — One-mark Questions

1. What is Phonetics? 1 mark CO2

The systematic study of speech sounds.

2. What is a phoneme? 1 mark CO2

The smallest sound unit that can distinguish meaning.

3. What is a minimal pair? 1 mark CO2

Two words differing in one sound, such as ship/sheep.

4. What is word stress? 1 mark CO2

Extra prominence given to one syllable in a word.

5. What is intonation? 1 mark CO2

The rise and fall of voice pitch in speech.

6. What is a pause? 1 mark CO2

A brief break that separates ideas and improves meaning.

7. Which punctuation mark ends a direct question? 1 mark CO2

Question mark.

8. When is a capital letter used? 1 mark CO2

At sentence beginnings and for proper nouns, among other uses.

9. What is self-introduction? 1 mark CO2

A brief organised presentation of one's identity, background, skills and purpose.

10. What is note making? 1 mark C01

Organising key information in concise headings, keywords and symbols.

11. What is the difference between note making and note taking? 1 mark C01

Note taking records information during input; note making reorganises it meaningfully afterwards.

12. What is listening for gist? 1 mark C01

Listening to understand the general idea.

13. What is listening for detail? 1 mark C01

Listening to locate specific facts or instructions.

14. Name one falling-intonation context. 1 mark C02

Statements or WH-questions.

15. Name one rising-intonation context.

1 mark C02

Yes/no questions.

Module III – Three-mark Questions

1. Differentiate vowels and consonants.

3 marks C02

Vowels are produced with relatively open airflow; consonants involve partial or complete obstruction at a place of articulation.

2. Explain minimal pairs with examples.

3 marks C02

Minimal pairs differ in one phoneme and show that the sound changes meaning, e.g., ship/sheep, fan/van and tin/ten.

3. Explain word stress and sentence stress. 3 marks C02

Word stress highlights one syllable; sentence stress highlights important content words and helps convey focus.

4. State three functions of intonation.

3 marks C02

It distinguishes question/statement patterns, expresses attitude and highlights information structure.

5. Correct the punctuation: 'after testing the motor rina said it is safe'

3 marks C02

After testing the motor, Rina said, "It is safe."

6. Write a three-part self-introduction plan. 3 marks C02

Opening identity;
education/skills/experience; goal and polite closing.

7. Explain listening for gist, specific information and inference. 3 marks C01

Gist finds overall message; specific listening extracts facts; inference identifies implied attitude or purpose.

8. Write three effective note-making abbreviations or symbols. 3 marks C01

e.g., approx. = approximately, → = leads to, + / – = advantage/disadvantage.

9. Convert a paragraph into notes.

3 marks C01

Use a title, numbered headings, subpoints, keywords and a key for abbreviations; avoid full sentences.

10. Explain the role of pauses in reading aloud. 3 marks C02

Pauses mark punctuation and idea groups, prevent ambiguity, support breathing and improve listener comprehension.

Module III — Seven-mark Questions

1. Explain English speech sounds at diploma level. 7 marks C02

Cover vowels/consonants, phonemes, selected IPA symbols, voiced/voiceless contrast, minimal pairs and pronunciation practice.

2. Explain stress, intonation and pauses in effective reading. 7 marks C02

Define each, show notation, discuss meaning and attitude, and demonstrate on a short passage.

3. Prepare a professional self-introduction for a job interview. 7 marks C02

Include greeting, name/place, qualification, technical interests, strengths with evidence, goal and thanks; keep it under 90 seconds.

4. Listen to a safety briefing and prepare structured notes. 7 marks C01

Expected: title, purpose, hazards, required PPE, step sequence, emergency instruction and concise symbols/abbreviations.

5. Explain punctuation and capitalization with examples. 7 marks C02

Cover full stop, comma, colon, semicolon, question/exclamation marks, apostrophe, quotation marks and major capitalization rules.

6. Design a listening-comprehension strategy for technical instructions. 7 marks C01

7 marks C01

Preview context, listen for sequence words, record measurements/warnings, verify unfamiliar terms, paraphrase instructions and confirm by feedback.

Module III — MCQ Practice

1. The smallest contrastive sound unit is a _____. MCQ C02

Phoneme.

2. ship/sheep is an example of a _____. MCQ C02

Minimal pair.

3. A yes/no question commonly uses _____ intonation. MCQ C02

Rising.

4. Important content words usually receive _____. MCQ C02

Sentence stress.

5. A brief organised personal presentation is a ____. MCQ C02

Self-introduction.

6. The symbol '→' in notes means ____. MCQ C01

leads to / results in.

7. A proper noun begins with a ____. MCQ C02

Capital letter.

8. Listening for the overall message is listening for ____. MCQ C01

Gist.

9. A colon can introduce a ____. MCQ C02

List or explanation.

10. Apostrophes can show possession or ____. MCQ C02

Omission in contractions.

Module IV – Writing

1. Translation between English and Mother Tongue

Understand complete meaning →

Identify tone and context →

Protect technical terms →

Reconstruct naturally →

Compare with source →

Proofread

English → Malayalam

Source: Switch off the main supply before opening the panel.

Target: Panel ഏകദേശം പ്രധാന വിദ്യുത് സപ്ലൈ സ്വിച്ച് ഓഫ് ചെയ്യുക.

Malayalam → English

Source: Reading ഏകദേശം പ്രധാന വിദ്യുത് സപ്ലൈ സ്വിച്ച് ഓഫ് ചെയ്യുക.

Target: If the reading is not stable, the test must be repeated.

Do not translate word by word. Preserve purpose, safety meaning, politeness, tense and technical terminology. Review machine translation carefully.

Malayalam explanation: Translation-ഏകദേശം പ്രധാന വിദ്യുത് സപ്ലൈ സ്വിച്ച് ഓഫ് ചെയ്യുക. Complete meaning, context, tone, technical accuracy ഏകദേശം target language-ഏകദേശം natural ഏകദേശം.

2. Effective and Responsible Use of AI Tools

Useful applications

- Brainstorming and outlining
- Grammar and clarity suggestions
- Summarising supplied text
- Drafting schedules and checklists
- Translation support
- Presentation structure

Mandatory human checks

- Verify facts, numbers and sources
- Check technical terminology
- Remove confidential information
- Review bias and tone
- Rewrite in your own voice
- Follow institutional disclosure rules

A practical prompt structure

ROLE + TASK + CONTEXT + AUDIENCE + CONSTRAINTS + REQUIRED FORMAT + CHECKS

Example: Act as a technical-writing assistant. Rewrite the supplied maintenance note as a professional email to the production supervisor. Keep all measurements unchanged, use a neutral tone, limit it to 120 words, add a clear subject line, and list any fact that needs verification.

Privacy: Never paste passwords, personal IDs, confidential drawings, customer information, proprietary reports or unpublished company data into public AI tools.

Malayalam explanation: AI output final answer ഏതു ഏതുവേണ്ടി എഴുതേണ്ടതെന്ന്. Facts, numbers, technical terms, privacy, bias എങ്ങനെ verify ചെയ്യാം judgement എങ്ങനെ എടുക്കാം edit ചെയ്യാം.

3. Professional Email

Clear subject →

Professional greeting →

Purpose in first line →

Necessary facts →

Required action/deadline →

Polite close →

Proofread

Subject: Request to Reschedule Laboratory Assessment

Dear Madam,

I am writing to request permission to attend the laboratory assessment on 18 August instead of 17 August because I have been assigned to an official industrial visit on that day. I have attached the visit circular for reference. Kindly let me know whether the alternative date is acceptable.

Yours sincerely,

Arun P.

Semester I, Electrical Engineering

Email checklist

- Does the subject state the purpose?
- Is the request or action clear?
- Are dates, names and attachments correct?
- Is the tone professional and concise?

4. Formal Letters and Meeting Invitations

Complaint letter structure

1. Reference and factual problem

2. Date/product/service details
3. Impact and evidence
4. Requested remedy
5. Reasonable response period
6. Polite close

Meeting invitation

- Purpose and agenda
- Date and time
- Venue or meeting link
- Participants
- Required preparation
- RSVP/contact

Subject: Review Meeting — Prototype Test Results

You are invited to a review meeting on 22 August at 2:00 p.m. in Lab 2 to discuss the prototype test results and corrective actions. Please bring the completed observation sheet and be prepared to explain any abnormal reading. Kindly confirm attendance by 20 August.

5. Technical and Professional Reports

Section	Purpose
Title / Identification	States what, where and when.
Objective / Scope	Defines purpose and limits.
Method / Procedure	Explains what was done.
Observations / Findings	Presents verified facts, readings and evidence.
Discussion	Interprets findings without hiding uncertainty.
Conclusion	Answers the objective.
Recommendation	States practical next action.

Short incident report

Date/Location: 12 July, Electrical Laboratory

Incident: During a no-load motor test, the current reading fluctuated between 1.2 A and 2.1 A.

Immediate action: The supply was isolated and the test was stopped.

Finding: A loose meter lead was identified after visual inspection with power removed.

Corrective action: The lead was secured, continuity was verified and the test was repeated under supervision.

Result: The current remained stable at 1.3 A.

Reporting rule: Separate observed facts from assumptions. Do not state a cause as certain

6. Presentation Planning and Delivery

Audience and purpose →

Opening →

2-3 key points →

Evidence/visual →

Transitions →

Conclusion →

Questions

Slide design

- One clear message per slide
- Large readable text
- Visual evidence instead of decoration
- Consistent layout
- Source any borrowed material

Delivery

- Practise timing
- Use eye contact and natural gestures
- Speak in idea groups
- Do not read every word from slides
- Close with the main takeaway

Two-minute presentation outline: Why Preventive Maintenance Matters

1. Hook: A small unchecked fault can stop an entire system.
2. Point 1: Planned inspection identifies wear early.
3. Point 2: Maintenance improves safety and reliability.
4. Point 3: Records support better decisions.
5. Conclusion: Preventive maintenance reduces risk, downtime and cost.

Malayalam explanation: Presentation- slide ഏകദേശം 2-3 മിനുട്ട് delivery ചെയ്യണം. Audience-ക്ക് വ്യക്തമായ clear structure, suitable pace, stress, pauses, supporting visuals ഉൾപ്പെടെ ഉപയോഗിക്കണം.

Module IV – One-mark Questions

1. What is Translation? 1 mark C03

Expressing the meaning of a source text accurately and naturally in another language.

2. What is the source language? 1 mark C03

The language of the original text.

3. What is the target language? 1 mark

C03

The language into which the text is translated.

4. Name one risk of word-for-word translation. 1 mark C03

It may produce unnatural or incorrect meaning.

5. What is an AI writing assistant? 1 mark

C04

A software tool that can generate, revise or organise text from instructions.

6. Name one rule for responsible AI use.

1 mark C04

Verify facts and edit the output before use.

7. What is a prompt? 1 mark C04

An instruction or input given to an AI tool.

8. What should never be entered into a public AI tool? 1 mark C04

Sensitive personal, confidential or proprietary information.

9. State one purpose of a professional email subject line. 1 mark C03

It tells the recipient the message purpose quickly.

10. What is a formal salutation? 1 mark

C03

A professional greeting such as Dear Sir/Madam or Dear Ms. Nair.

11. Name one feature of a technical report. 1 mark C03

Clear structure with objective evidence.

12. What is a complaint letter? 1 mark C03

A formal message that states a problem and requests a reasonable remedy.

13. What is a meeting invitation? 1 mark

C03

A message specifying purpose, date, time, place/link and expected participants.

14. What is a presentation signpost?

1 mark C03

A phrase that guides the audience, e.g., 'Let us move to the next point.'

15. Why must AI-generated translation be reviewed? 1 mark C04

It may miss context, tone, terminology or cultural meaning.

Module IV — Three-mark Questions

1. State three principles of good Translation. 3 marks C03

Preserve meaning, use natural target-language expression and maintain suitable tone/technical terminology.

2. Explain three responsible uses of AI tools. 3 marks C04

Brainstorming, language revision and summarising are acceptable when the user verifies, edits and acknowledges assistance as required.

3. Write a professional email subject and opening. 3 marks C03

Subject: Request for Industrial Visit Permission. Dear Sir, I am writing to request permission for...

4. State three parts of a formal letter. 3 marks C03

Sender/date and recipient details; salutation/subject/body; complimentary close and signature.

5. Explain the 3C principle of professional writing. 3 marks C03

Clear: easy to understand; Concise: no unnecessary words; Correct: accurate language and facts.

6. Write three sections of a short technical report. 3 marks C03

Purpose/Introduction, Findings/Discussion and Conclusion/Recommendation.

7. Explain how AI can support scheduling and summarising. 3 marks C04

AI can convert requirements into a draft schedule and condense long text, but dates, priorities, omissions and confidentiality must be checked.

8. Draft a concise meeting invitation. 3 marks C03

Include topic, date/time, venue or link, agenda, required preparation and RSVP/contact.

9. Write three presentation delivery tips. 3 marks C03

Use clear structure, maintain eye contact and pace, and support points with readable visuals rather than dense slides.

10. Differentiate proofreading and editing. 3 marks C04

Proofreading corrects surface errors; editing improves content, structure, clarity and coherence.

Module IV — Seven-mark Questions

1. Translate a short workplace message between English and Malayalam and explain your choices. 7 marks C03

A strong answer preserves purpose, politeness and technical terms, avoids word-for-word distortion, and comments on any necessary reordering or contextual wording.

2. Explain a responsible workflow for using AI writing assistants. 7 marks C04

Define task and audience; provide context; avoid sensitive data; request structured output; verify facts; check bias/terminology; rewrite in own voice; disclose use when required.

3. Draft a professional email responding to a job advertisement. 7 marks C03

Include meaningful subject, formal greeting, reference to vacancy, qualification and relevant skills, attachment mention, request for consideration and professional closing.

4. Prepare a formal complaint letter about defective equipment. 7 marks C03

Include facts, purchase/service details, defect and impact, evidence, requested remedy, reasonable deadline and polite closing.

5. Write a technical incident report.

7 marks C03

Use title/date/location, objective description, people/equipment involved, sequence, immediate action, cause if verified, recommendation and signature.

6. Plan a two-minute technical presentation. 7 marks C03

Provide opening hook and objective, 2–3 main points, one visual/example, transitions, conclusion and delivery plan.

Module IV — MCQ Practice

1. The original language is the _____. MCQ C03

Source language.

2. The best Translation preserves _____. MCQ C03

Meaning and natural expression.

3. Confidential data should be ____ from public AI tools. MCQ C04

Kept out.

4. A professional email needs a clear _____. MCQ C03

Subject line.

5. AI output must be _____. MCQ C04

Verified and edited.

6. A report should distinguish fact from _____. MCQ C03

Opinion or assumption.

7. The final part of a presentation should _____. MCQ C03

Summarise and close.

8. A complaint letter should request a reasonable _____. MCQ C03

Remedy.

9. A prompt should state task, context and _____. MCQ C04

Expected output.

10. Proofreading mainly checks _____. MCQ C04

Surface errors such as spelling, punctuation and grammar.

LANGUAGE REFERENCE BANK

Compact rules, templates and checklists

3C Writing Rule

Clear: one unambiguous meaning.

Concise: no unnecessary words.

Correct: accurate grammar, facts and format.

Paragraph Structure

Topic sentence → supporting details/examples → logical connectors → concluding sentence.

Professional Tone

Specific, respectful, neutral and action-oriented. Avoid slang, blame and emotional exaggeration.

Summary Formula

Main idea + essential supporting points + logical order – examples/repetition/opinion.

Process Formula

Purpose + inputs + ordered steps + checks/safety + result.

AI Review Formula

Generate → verify → compare with source → edit → protect privacy → take responsibility.

Useful Linking Words

Function	Words/Phrases
Addition	also, furthermore, in addition
Contrast	however, whereas, in contrast
Cause	because, since, due to
Effect	therefore, consequently, as a result
Sequence	first, next, subsequently, finally
Example	for example, such as, for instance
Conclusion	in conclusion, therefore, to summarise

Commonly Confused Words

Pair	Difference
accept / except	receive / excluding
affect / effect	usually verb influence / usually noun result
advice / advise	noun / verb
principal / principle	main/person in charge / rule
stationary / stationery	not moving / writing materials
ensure / insure	make certain / provide insurance

SELF-LEARNING CENTRE

Official 15-week activity plan

Each activity includes a concrete output. The official PDF labels Weeks 13–15 under “Module I”; their content belongs to Writing, so this handbook groups them under Module IV and records the source-label issue transparently.

Week 1 · 4 h

Week 2 · 3 h

Listen to/read a famous English speech and summarise the main points in one paragraph.

Output: source details + 100-word summary.

Week 3 · 4 h

Construct an effective conversation for a real-life situation.

Output: 12–16 line dialogue.

Week 5 · 4 h

Treat an English song or authentic translation as poetry and analyse devices.

Copyright note: Do not reproduce full lyrics; quote only short necessary phrases.

Week 7 · 4 h

Find Passive Voice in a reputed article and rewrite five sentences in Active Voice.

Output: source citation + transformations.

Week 9 · 3 h

Identify twenty words and write phonetic transcriptions.

Week 11 · 5 h

Listen to suitable stand-up comedy and write a response in fewer than 100 words.

Output: topic, technique, viewpoint and language note.

Week 13 · 5 h

Translate a mother-tongue diary entry into English and an English paragraph into the

Identify ten words and form new words using prefixes and suffixes.

Output: root, affix, new word, meaning, sentence.

Week 4 · 4 h

Correct subject-verb agreement errors.

Output: 20 corrected sentences with rule notes.

Week 6 · 4 h

Describe what you are doing now, then convert it into Past and Future Tense.

Week 8 · 3 h

Convert a celebrity interview's direct questions and answers into Reported Speech.

Week 10 · 4 h

Read a newspaper article and summarise it in exactly 50 words.

Week 12 · 3 h

Prepare and deliver a job-interview self-introduction.

Week 14 · 5 h

Draft a job-response email, complaint letter or meeting invitation.

mother tongue.

Week 15 · 5 h

Create and deliver a two-minute presentation on any suitable topic.

Total official self-learning hours: 60.

ASSESSMENT AND MODEL PAPERS

Evaluation criteria and examination preparation

Practical Continuous Evaluation

Criterion	Description	Marks
Preparation and Punctuality	Attendance and timely completion	5
Learning Experience and Application	Apply language in a new context	10
Passive Skills	Listening and Reading mastery	10
Active Skills	Speaking and Writing ability	10
Viva / Concept Understanding	Organisation and clarity	10
Interpersonal Communication	Language use in varied contexts	5
Total		50

Practical Test Scheme

Category	Focus	Marks
Listening and Reading	Comprehend, evaluate, appreciate; correct stress and intonation	10
Speaking and Writing	Description, real-life language, pronunciation	10
Translation	Translate short paragraphs and understand language variation	10
AI Tools	Apply AI tools for content creation and communication	5
Grammar Application	Accurate real-life language use	5
Total		40

End Semester Theory Pattern

Duration: 2.5 hours · Total: 60 marks.

Per module	Part A	Part B	Part C	Total
Modules I-IV	2 questions × 1 = 2	Answer 2 of 3 × 3 = 6	1 set with choice × 7 = 7	15 per module

Practice Model Paper — Not an Official SITTTR Question Paper

Course: 1001 English for Technical Communication · **Duration:** 2.5 hours · **Marks:** 60

Part A — Answer all questions (8 × 1 = 8)

1. Define scanning.
2. Give one example of a suffix.
3. What is imagery?
4. Change to Passive Voice: The team completed the test.
5. What is word stress?
6. Name one use of a colon.
7. What is target language?
8. Name one responsible AI practice.

Part B — Answer two of three from each module (8 × 3 = 24)

9. Differentiate skimming and scanning.
10. Explain three concord rules.
11. Write a workplace permission dialogue.
12. Explain any three poetic devices.
13. Convert three sentences from Active to Passive Voice.
14. State three Reported Speech changes.
15. Explain word stress, sentence stress and intonation.
16. Correct punctuation in a supplied paragraph.
17. Explain three note-making techniques.
18. State three principles of Translation.
19. Write a professional email opening.
20. Explain responsible AI verification.

Part C — Answer one from each set (4 × 7 = 28)

21. **Module I:** Summarise a supplied technical passage in 80–100 words *OR* explain sentence structure, articles, auxiliaries and concord.
22. **Module II:** Appreciate the poem “Blueprint at Dawn” *OR* explain Active/Passive Voice and Reported Speech with examples.
23. **Module III:** Prepare a professional self-introduction *OR* explain Phonetics, stress, intonation and pauses.
24. **Module IV:** Draft a job-response email *OR* explain a responsible workflow for AI-assisted professional writing.

Model Answer Guidance

Part A: Direct definition or transformation.

Part B: Three distinct explained points or rules with examples.

Part C: Organised introduction, complete core points, accurate examples/template, and a brief conclusion. Use headings where suitable.

Series-test source note: The official series-examination table is formatted ambiguously. This handbook reproduces the unambiguous End Semester pattern for the full model paper and does not invent missing series-test interpretation.

QUICK REVISION

One-day revision and examination checklist

Module I

- Main idea vs details
- Summary steps
- Word formation
- Sentence patterns
- Articles, auxiliaries, concord
- Situational expressions

Module II

- Theme/tone/devices
- Tense forms
- Active–Passive
- Description structures
- Reported Speech

Module III

- Speech sounds
- Stress/intonation/pauses
- Punctuation
- Self-introduction
- Listening and notes

Module IV

- Translation principles
- AI verification/privacy
- Email and letter
- Report structure
- Presentation delivery

Writing check

1. Have I answered the exact task?
2. Is the format correct?
3. Are ideas logically ordered?
4. Are grammar and punctuation accurate?
5. Are names, dates and technical terms verified?

Speaking check

1. Clear opening and purpose

2. Correct pronunciation of key terms
3. Stress important words
4. Use meaningful pauses
5. Conclude confidently

Syllabus coverage: All official major topics, detailed learning outcomes, 15 self-learning activities, assessment criteria and ESE pattern from the supplied 2026 Course 1001 PDF are represented in this handbook.

REFERENCES AND SOURCE DECLARATION

Learning resources

Official textbook

Course Book prepared by SITTTTR.

Selected references

- Stephen Fry — *The Ode Less Travelled*
- Kim Addonizio and Dorianne Laux — *The Poet's Companion*
- Meenakshi Raman and Sangeeta Sharma — *Technical Communication: Principles and Practice*
- J. D. O'Connor — *Better English Pronunciation*
- M. Ashraf Rizvi — *Effective Technical Communication*

Web resources listed in the syllabus

- British Council LearnEnglish
- NPTEL communication course
- Cambridge English
- PoemHunter
- EnglishClub
- SlideShare

Official source declaration

Academic scope, course identity, outcomes, modules, activities and assessment structure were derived from the supplied SITTTTR Kerala Diploma Curriculum Revision 2026 Course 1001 syllabus PDF. Explanations, practice passages, poem, examples and questions in this handbook are original educational material and are not presented as official SITTTTR course-book text or official question papers.